



Mayfly taxonomic and functional diversity in hydrologically extreme habitats of temporary pools in the Mediterranean karst intermittent rivers

Marina Vilenica¹ · Fran Rebrina² · Lea Ružanović² · Mario Rumišek² · Renata Matoničkin Kepčija² · Andreja Brigić²

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Abstract

Temporary pools are small and shallow freshwater habitats with a recurring dry phase and intermittent stagnant water. They are particularly well represented in semiarid and karstic regions of the world, where they form during the lentic phase of intermittent rivers and streams. Despite the increasing research interest in such habitats, community ecology studies are still rare, and thus far, have mostly focused on the lotic hydrological phase. Mayflies are among the most abundant benthic macroinvertebrates in freshwater ecosystems, showing high sensitivity to habitat degradation, which makes them widely used in bio-monitoring programs. Mayfly taxonomic and functional diversity, as well as their relationship with environmental variables, were investigated in the temporary pools formed during the lentic phase of three karst intermittent rivers in the Mediterranean. Pools were characterized by river-specific species composition. However, mayfly taxonomic and functional diversity metrics were comparable among the pools of the three rivers. They were influenced by distance from the river's source and physicochemical water properties, particularly water temperature, conductivity, chemical oxygen demand, and nitrate concentration. Those are related to habitat morphology, substrate composition, and the level of anthropogenic impact. Although the investigated small and shallow temporary pools represent rather harsh habitats, they were inhabited by some species rare in Croatian freshwater habitats (*Procladius pennulatus* and *Nigrobaetis niger*). Our findings suggest that the conservation value of Mediterranean intermittent rivers does not rest upon a single hydrological phase, but upon an inter-change of different phases.

Keywords Ephemeroptera · Species richness · Functional traits · Temporary habitats · Lentic habitats · Environmental variables

Introduction

Temporary pools are small, ever-changing freshwater ecosystems with stagnant water and a highly variable hydroperiod duration. They include habitats ranging from small puddles where water retains for only a few days after rainfall,

to more permanent ones characterized by water cessation lasting only several weeks in a year (Blaustein and Schwartz 2001; Williams et al. 2001). Connected or isolated temporary pools also often form during the lentic phase of intermittent rivers and streams (Larned et al. 2010; Detry et al. 2017). Although such habitats occur in various climate regions of the world (Williams et al. 2001), they are particularly well represented in semiarid and karstic regions where dynamic ecosystems of intermittent rivers are common (Larned et al. 2010). Once inundated, temporary pools retain water long enough for some organisms to be able to complete the aquatic phase of their life cycle (Blaustein and Schwartz 2001). They are often considered biodiversity hotspots, possibly characterized by specialized assemblages encompassing rare and endemic species (e.g., Baskin 1994; Zacharias et al. 2007). However, due to their small size, such

✉ Marina Vilenica
marina.vilenica@ufzg.hr

✉ Andreja Brigić
andreja.brigic@biol.pmf.hr

¹ Faculty of Teacher Education, University of Zagreb, Trg Matice Hrvatske 12, Petrinja, Croatia

² Faculty of Science, Department of Biology, University of Zagreb, Horvátovac 102a, Zagreb, Croatia

habitats are highly susceptible to pollution and the effects of climate change, and are thus highly threatened worldwide (Williams et al. 2001).

Over the past decades, research interest in intermittent rivers and streams has increased (e.g., Leigh et al. 2016; Datry et al. 2017; Borg Galea et al. 2019), resulting in a better understanding of the diverse and unique biota they harbor, their importance for the ecological integrity of river networks, and the ecosystems services they provide (Steward et al. 2012; Datry et al. 2018). Hence, recent research efforts have provided a satisfactory basic knowledge about the communities inhabiting intermittent rivers during the lotic phase (Williams 2006; Datry et al. 2014; Schriever and Lytle 2016; Stubbington et al. 2017a, b; Vilenica et al. 2023). However, ecological studies of aquatic biota occurring in the pools formed during the lentic hydrological phase are still rather scarce (Schriever and Lytle 2016; Hill and Milner 2018). Since temporary pools represent the least studied but nevertheless integral phase of the intermittent river cycle, it is essential to thoroughly evaluate their biotic diversity. To better understand the posing threats and provide tools to protect these vulnerable ecosystems, it is crucial to determine if, how, and why temporary pool communities differ among different rivers.

Freshwater macroinvertebrates are an important component of biotic diversity and have several important ecological roles in various aquatic habitats, including intermittent rivers and their temporary pools (Stubbington et al. 2017a, b). Some of them are crucial for leaf litter processing (Richardson 1990), while others have a vital role in secondary production, representing a substantial food resource for both aquatic (Courtwright and May 2013) and terrestrial predators (Lynch et al. 2002). Intermittent rivers are considered harsh habitats from the aquatic communities' perspective, as macroinvertebrates are challenged with high oscillations in environmental conditions that result from the alteration of the lotic, lentic, and dry hydrological phases (Boulton 2003). To survive the dry period, they often rely on life history traits related to resilience (capacity to recolonize and reestablish populations in newly established lotic phase after the water flow returns) and resistance (capability to survive the dry riverbed phase due to biological adaptations or in refugia)-based strategies (Bogan et al. 2015; Datry et al. 2016). To survive through flow cessation, aquatic macroinvertebrates can utilize different refuges: nearby perennial lotic (Chester and Robson 2011) and lentic (Paltridge et al. 1997) habitats, instream pools (Boersma et al. 2014), moist microhabitats (woody debris, leaf litter, algal mats, and muddy sediment) (Boulton 1989; Robson et al. 2011), or the hyporheic zone of intermittent rivers (Stubbington 2012).

Mayflies are among the most abundant amphibiotic benthic macroinvertebrates found in various freshwater ecosystems (Elliott et al. 1988; Brittain and Sartori 2003;

Bauernfeind and Soldán 2012). Since many species show high sensitivity to habitat degradation, mayfly taxonomic and functional diversity and their relationship with environmental variables are used in biomonitoring programs worldwide (Lenat and Penrose 1996; Bauernfeind and Moog 2000; Ferro and Sites 2007; Espinosa et al. 2023). Functional diversity, or diversity of life history traits of species within an assemblage, is often used to understand the link between the taxonomic diversity and ecosystem function, to recognize ecological processes influencing biotic assemblages, and to evaluate impacts of habitat characteristics and anthropogenic habitat alterations on those assemblages (Villéger et al. 2008). Functional diversity reflects the degree of functional differences (number of functional groups) that exist among the species within an assemblage (Tilman 2001). Hence, studying the assemblage parameters (i.e., taxonomic and functional diversity) of freshwater macroinvertebrates, including mayflies, in temporary pools of intermittent rivers, could help to address the existing gap in our understanding of the biota of such habitats and their relationship with such extreme environments. Moreover, this knowledge could be used in planning conservation activities considering the effects of climate change that modify the hydrological regime of numerous freshwater habitats, where an increasing number of perennial regimes are becoming intermittent worldwide (Larned et al. 2010).

The main goals of this study were: (i) to assess the differences in mayfly assemblage metrics (abundance, species richness, diversity) and similarity in assemblage composition among the temporary pools of three Mediterranean intermittent rivers, (ii) to identify functional diversity of mayfly assemblages and detect changes among pools in their functional traits, and (iii) to determine key environmental variables shaping mayfly assemblages in the studied pools.

Material and methods

Study area

The study area is located in the Dinaric western Balkan ecoregion (ER5) in Croatia (Illies 1978). In this region, the climate is temperate humid with hot summers (Cfa, Köppen classification), with mean temperature of the warmest month above 22 °C (Šegota and Filipčić 2003), mean annual air temperature of 14 °C, and mean annual rainfall of about 1000 mm (Zaninović et al. 2008).

Three karst Mediterranean intermittent rivers of the Adriatic basin have been included in the study: Krčić, Čikola, and Miljašić Jaruga (Fig. 1), all of them characterized by a dry period lasting several months during summer (Bonacci 1985; 2014; Bonacci et al. 2006; 2019; Miliša et al. 2020; DHMZ 2023). Lotic phase of the rivers was sampled at three

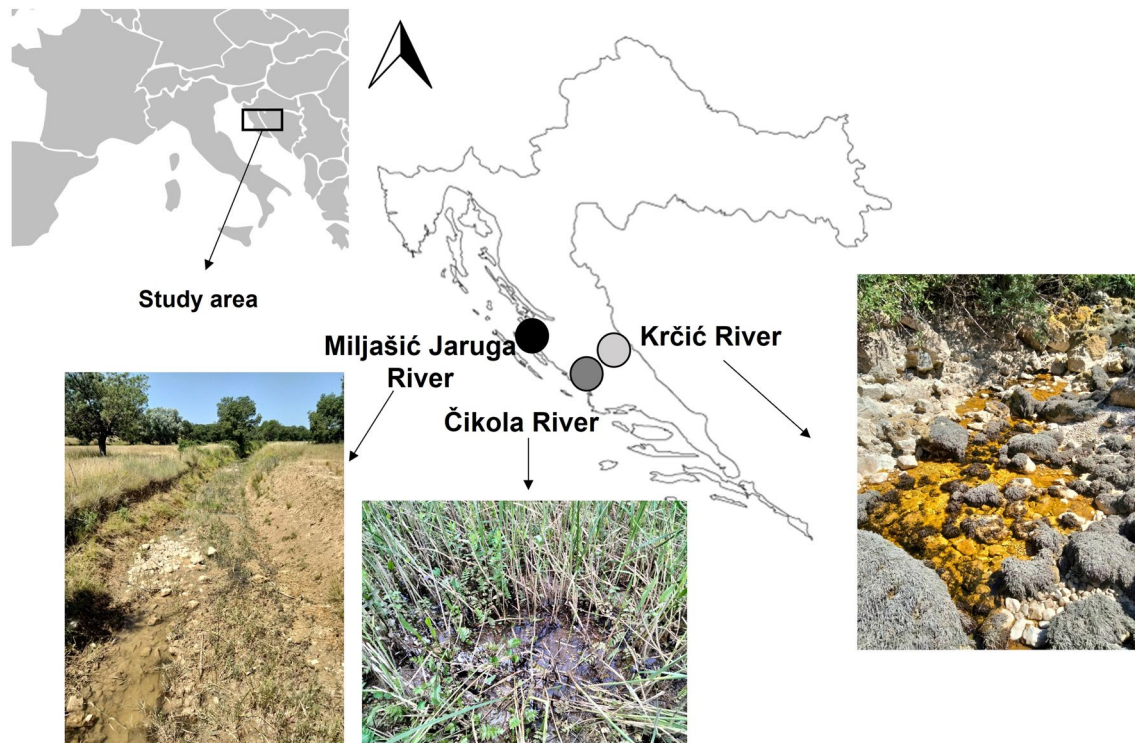


Fig. 1 Map showing the position of the study area in the karst Mediterranean region of Croatia, with locations of the studied intermittent rivers and photographs depicting temporary pools sampled at each intermittent river

Table 1 Distances between study sites and river source at three Mediterranean karst intermittent rivers (Croatia)

River	Study site	Distance from the source
Krčić	1	1118 m
	2	2203 m
	3	8725 m
Čikola	1	10,577 m
	2	19,105 m
	3	31,918 m
Miljašić Jaruga	1	11,358 m
	2	13,715 m
	3	16,026 m

study sites each, where study site 1 was the closest and study site 3 furthest from the river source (Vilenica et al. 2023). The distances between study sites and river source were calculated using QGIS 3.22.5 and are presented in Table 1.

Temporary pools were sampled during the lentic hydrological phase along the river course in line with the pools' formation period at the same study sites. A total of 23 temporary pools were sampled, depending on pool availability at each river. Therefore, eight pools were sampled in the Krčić and Čikola rivers (two pools at study site 1, and three at

study sites 2 and 3), and seven were sampled in the Miljašić Jaruga River (three pools at study sites 1 and 2 and one pool at study site 3).

Environmental variables and sampling

The study was conducted between June and August 2021, depending on the time of pool formation along the course of each river. Mayfly nymphs, along with other benthic invertebrates, were collected using a Surber sampler (25 cm × 25 cm, mesh size 500 µm). The samples were preserved in 70% ethanol and later sorted and identified in the laboratory using standard identification keys (Waltz et al. 1994; Bauernfeind and Humpesch 2010).

Simultaneously with benthic invertebrate sampling, the following environmental variables were measured in situ in each pool: water temperature, dissolved oxygen concentration, and saturation (using the WTW Oxi 330/SET oximeter), conductivity (using the WTW LF 330 conductivity meter), pH (using the WTW pH 330 pH-meter), pool width, pool length, and water depth (using a handheld meter). Additionally, we collected a 1-L water sample from each pool for the laboratory analysis of water chemical properties (chemical oxygen demand, nitrite, nitrate, and orthophosphate concentrations) using standard analytical procedures (Greenberg 1992).

Data analyses

To compare mayfly assemblage metrics (abundance, species richness, Shannon diversity index) among the pools of the three investigated intermittent rivers, one-way analysis of variance (ANOVA) or Kruskal–Wallis H tests were used depending on data distribution, followed by Unequal N HSD or multiple comparisons post hoc tests, respectively. Data were tested for normality using Shapiro–Wilk W test and transformed if necessary (sqrt, 4th root, log). The same tests were applied to detect differences in the environmental variables measured in the pools of the three investigated intermittent rivers. The above-mentioned analyses were performed in Statistica 13.0 (Statsoft 2010).

The similarities in composition of mayfly assemblages between the pools of three intermittent karst rivers were assessed using the non-metric multidimensional scaling analysis (NMDS) based on the Bray–Curtis similarity matrix. Prior to analysis, taxa abundance data were $\log(x + 1)$ transformed. Replicates without mayfly records were excluded from the analysis. Similarity percentage (SIMPER) analysis was used to assess which mayfly species were primarily responsible for the similarities between the pools of the investigated rivers and which species contributed to similarities among the rivers. Bray–Curtis similarity index and Shannon diversity index (H') calculation, NMDS, and SIMPER analyses were conducted in the Primer 7.0 software package (Clarke and Gorley 2006).

In total, 13 functional traits from four trait groups were used to quantify the functional diversity of mayfly assemblages in studied temporary pools: (i) functional feeding groups (grazers—feed on algae, biofilm, partially on particulate organic matter, and tissues of living plants; gatherers—feed on sedimented fine particulate organic matter), (ii) current preference (limnophil—preferably inhabit lentic habitats, can rarely be found in slowly flowing streams; rheophil to limnophil—preferably inhabit slowly flowing streams or lentic zones of lotic habitats, can also occur in lentic habitats; rheophil—inhabit lotic habitats with moderate-to-high water velocity); (iii) lateral connectivity preference (lotic habitats: eupotamon—main channel and connected side arms; parapotamon—side arms connected only at the downstream end at mean water levels; and lentic habitats, without connectivity to the main channel at mean water levels: plesiopotamon—coverage by macrophytes does not exceed 20%; palaeopotamon—coverage by macrophytes exceeds 20%; temporary water bodies—temporary pools, with the water level being mainly dependent on ground water levels); and (iv) life cycle type (bivoltine—with two generations per year; trivoltine—with three generations per year; flexible life cycle—with flexible number of life cycles per year) (extracted from Amoroso

et al. 1987; Schmedtje and Colling 1996; Buffagni et al. 2009, 2023; Schmidt-Kloiber and Hering 2015).

The Rao quadratic diversity (RaoQ) coefficient was used to quantify the functional diversity of mayfly assemblages in pools of the lentic phase of three intermittent rivers. RaoQ is a measure of trait convergence or divergence patterns compared with random expectation. Community weighted mean (CWM) values were calculated for each mayfly functional trait. The aim was to quantify shifts in mean trait values within the assemblages, resulting from environmental selection for certain functional trait categories (Ricotta and Moretti 2011). RaoQ and CWM values were calculated in CANOCO version 5.15 package (ter Braak and Šmilauer 2012). Before the analysis, normality of the functional data was tested using the Shapiro–Wilk test in Statistica 13.0 (Statsoft 2010). Differences in the RaoQ coefficient and trait CWM values among the pools of the three intermittent rivers were tested using the Kruskal–Wallis H test, followed by multiple comparison post hoc test in Statistica 13.0 (Statsoft 2010).

Canonical correspondence analysis (CCA) was used to assess the relationship between mayfly occurrence and environmental variables in the investigated pools. Redundancy analysis (RDA) was used to investigate the influence of environmental factors on the spatial distribution of CWM values of the four groups of functional traits in mayfly assemblages. The CCA and RDA analyses included all eight mayfly species (rare species were downweighed using the downweighing of rare species feature in the analyses) and seven environmental variables, which differed significantly among the rivers. Prior to the CCA, mayfly abundances were $\log(x + 1)$ transformed, while prior to the RDA, they were centered and standardized by the average functional traits. Monte Carlo permutation test with 999 permutations was applied to test the statistical significance of the relationships between species or trait composition and environmental variables. The analyses were performed in the CANOCO package version 5.15 (Ter Braak and Šmilauer 2012).

Spearman's correlation coefficient was used to analyze the relationship between the mayfly assemblage metrics and distance from the river source in the pools along the course of each studied intermittent river. The analysis was performed in Statistica 13.0 (Statsoft 2010).

Results

Environmental variables

The investigated pools significantly differed in water depth, water temperature, conductivity, pH, chemical oxygen demand, and concentration of nitrates and nitrites (Table 2; Fig. 2). The pools in the Miljašić Jaruga River

Table 2 Differences in physicochemical water properties among the pools of the three Mediterranean karst intermittent rivers (Croatia) (*One-way ANOVA or Kruskal–Wallis H test, d.f.* (degrees of freedom) = 2). Significant results ($p < 0.05$) are reported in bold

Abiotic parameter	Statistical test	<i>p</i> -Value	Comparison
Conductivity	$H = 16.71$	$p < 0.001$	Miljašić Jaruga, Čikola > Krčić
pH	$F = 10.09$	$p < 0.001$	Krčić > Miljašić Jaruga, Čikola
Nitrate concentration	$F = 39.95$	$p < 0.001$	Miljašić Jaruga > Krčić, Čikola
Water temperature	$H = 11.43$	$p < 0.01$	Miljašić Jaruga > Krčić
Chemical oxygen demand	$F = 9.01$	$p < 0.01$	Čikola > Krčić, Miljašić Jaruga
Water depth	$H = 6.37$	$p < 0.05$	Miljašić Jaruga > Krčić
Nitrite concentration	$F = 5.03$	$p < 0.05$	Miljašić Jaruga > Krčić, Čikola
Pool width	$F = 1.92$	$p > 0.05$	
Pool length	$F = 2.58$	$p > 0.05$	
Oxygen saturation	$F = 1.08$	$p > 0.05$	
Dissolved oxygen concentration	$F = 0.62$	$p > 0.05$	
Orthophosphate concentration	$F = 0.41$	$p > 0.05$	

had significantly lower water depth and higher water temperature compared with those in the Krčić River. The Krčić River pools had significantly lower conductivity and higher pH compared with those in the Čikola and Miljašić Jaruga rivers. Furthermore, the pools in the Čikola River had significantly higher chemical oxygen demand compared with the pools in the Krčić and Miljašić Jaruga rivers. Finally, the pools in the Miljašić Jaruga River exhibited significantly higher concentrations of nitrates and nitrites compared with the pools in Krčić and Čikola rivers (Table 2; Fig. 2). Significant differences were not observed in terms of pool width, pool length, oxygen saturation, dissolved oxygen concentration, or in-water orthophosphate concentration (Table 2; Fig. 2).

Mayfly assemblages and their functional diversity

Mayfly assemblage metrics, i.e., abundance, species richness, and Shannon diversity, did not differ significantly among the pools of the studied intermittent rivers (Table 3; Fig. 3). A total of eight species was recorded, five of them in the pools of the Čikola and Miljašić Jaruga rivers and four species in the pools of the Krčić River (Table 4). The highest mayfly abundance was recorded in the pools of the Čikola River (1994.0 individuals/m²), followed by the Krčić (1462.0 individuals/m²) and Miljašić Jaruga rivers (147.9 individuals/m²) (Table 4). The most abundant species in the study area was *Procladius pennulatus*, recorded in the pools of all studied rivers, followed by *Centroptilum luteolum* and *Cloeon dipterum* (Table 4). The least abundant species were *Baetis* cf. *bucuratus* and *Nigrobaetis niger*, recorded only in the pools of the Miljašić Jaruga and Čikola rivers, respectively (Table 4).

NMDS analysis showed low similarity (5%) in mayfly assemblages among the pools of the three intermittent rivers (Fig. 4). There is a separation of the pools belonging to different rivers, with the Krčić River pools being most distinct,

whereas the pools of the Miljašić Jaruga and Čikola rivers had more similar assemblages (Fig. 4; Table 4). Those two clusters share 15% similarity.

The SIMPER group similarity analysis (Table 5) showed that the pools of the Krčić River differed from the pools of other two rivers mainly due to the presence of *Baetis rhodani*. Furthermore, their assemblages lacked *Caenis* species recorded in the pools of the other two rivers, that is, *Caenis horaria* in the pools of the Čikola River and *Caenis luctuosa* in the pools of the Miljašić Jaruga River, which were also responsible for the main differences between the pools of the latter two rivers (Table 5).

Functional diversity (RaoQ) of mayfly assemblages was comparable among the temporary pools of three Mediterranean karst intermittent rivers (Table 3; Fig. 5).

In terms of mayfly current preference, abundances of limnophil species were comparable among the pools of the studied rivers, the pools of the Miljašić Jaruga River had higher share of rheophil to limnophil species compared with the pools of the Krčić River, while rheophil species were present only in the pools of the Krčić River (Table 6; Fig. 6a).

Regarding mayfly lateral connectivity preference, a significantly higher share of lentic species (i.e., preferring pleiopotamon and palaeopotamon) was recorded in the pools of the Miljašić Jaruga River compared with the pools of the Krčić River (Table 6; Fig. 6b). Higher shares of species typical for temporary waterbodies were recorded in the pools of the Čikola River compared with the pools of the Krčić River (Table 6; Fig. 6b). Abundances of species preferring lotic habitats (i.e., eutopotamon and parapotamon) were comparable among the pools of the three intermittent rivers (Table 6; Fig. 6b).

Trophic structure of mayfly assemblages differed significantly among the pools of the three intermittent rivers; a significantly higher share of gatherers and a lower share of grazers was recorded in the pools of the Miljašić Jaruga River compared with the pools of other two rivers,

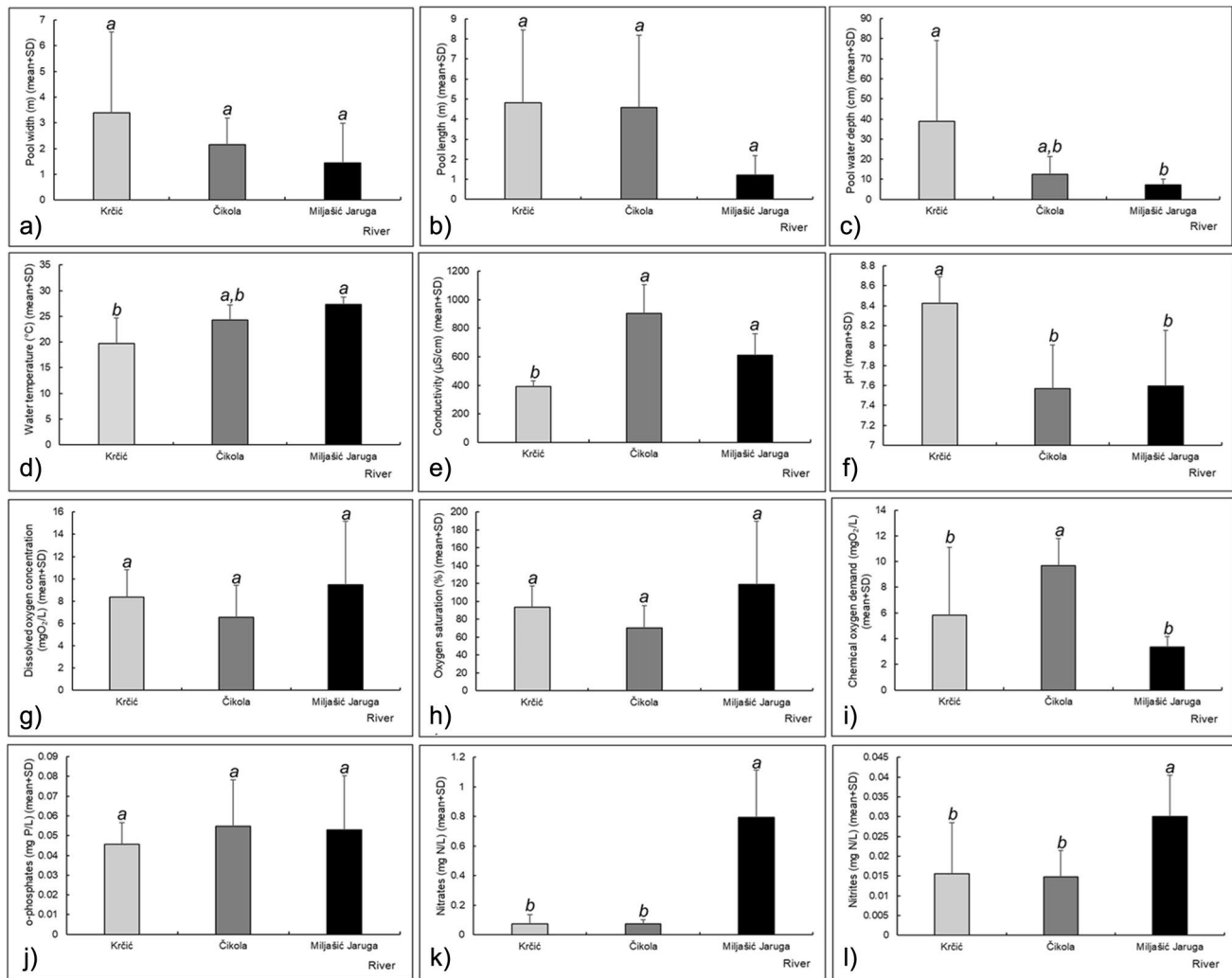


Fig. 2 Physicochemical water properties measured in the pools of the three Mediterranean karst intermittent rivers (Croatia), shown as mean with standard deviation (SD): **a** pool width, **b** pool length, **c** water depth, **d** water temperature, **e** conductivity, **f** pH, **g** dissolved oxygen concentration, **h** oxygen saturation, **i** chemical oxygen demand, **j** orthophosphate, **k** nitrate, **l** nitrite concentrations. Differ-

ent letters denote statistically significant differences among the pools of the three rivers (*One-way ANOVA* or *Kruskal–Wallis H test*, for significant results $p < 0.05$). Environmental variables were measured in eight pools in the Krčić and Čikola rivers and seven pools in the Miljašić Jaruga River

Table 3 Differences in mayfly assemblage parameters among the pools of the of the three Mediterranean karst intermittent rivers (Croatia) (*One-way ANOVA* or *Kruskal–Wallis H test*, $d.f. = 2$, $p > 0.05$)

Assemblage parameters	Statistical test	<i>p</i> -Value
<i>Taxonomic parameters</i>		
Abundance	$F = 2.01$	$p > 0.05$
Species richness	$H = 1.34$	$p > 0.05$
Shannon diversity	$H = 3.22$	$p > 0.05$
<i>Functional diversity parameter</i>		
RaoQ	$H = 3.97$	$p > 0.05$

where the proportions of feeding groups were similar (Table 6; Fig. 6c).

Pools of all three rivers had a comparable share of species with bivoltine life cycle type, trivoltine species were present only the pools of the Krčić River, while the pools of the Miljašić Jaruga River had significantly higher share of species with flexible life cycle compared with the pools of the Krčić River (Table 6; Fig. 6d).

Mayflies and environmental variables

The ordination of the mayfly species and environmental data included in the CCA is shown in the $F1 \times F2$ ordination plot (Fig. 7). The eigenvalues for the first two CCA

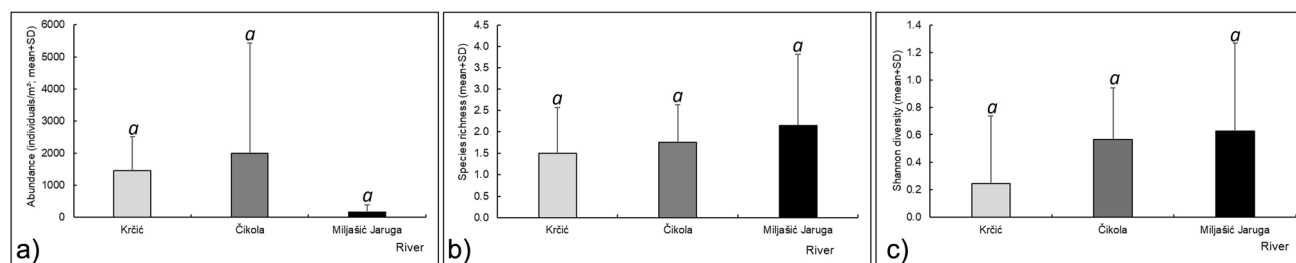


Fig. 3 Mayfly assemblage metrics in the pools of the lentic phase of three Mediterranean karst intermittent rivers (Croatia), shown as mean with standard deviation (SD): **a** species richness, **b** abundance

Table 4 Abundance (shown as mean number of individuals per m²) of mayfly species recorded in the pools of the lentic phase in three Mediterranean karst intermittent rivers (Croatia)

Mayfly species/river	Krčić	Čikola	Miljašić Jaruga
<i>Baetis cf. buceratus</i> Eaton, 1870			4.0
<i>Baetis rhodani</i> (Pictet, 1843)	746.0		
<i>Centropilum luteolum</i> Müller, 1776	10.0	774.0	17.9
<i>Cloeon dipterum</i> (Linnaeus, 1761)	702.0	4.0	14.0
<i>Nigrobaetis niger</i> (Linnaeus, 1761)		4.0	
<i>Procloeon pennulatum</i> (Eaton, 1870)	4.0	1206.0	2.0
<i>Caenis horaria</i> (Linnaeus, 1758)		6.0	
<i>Caenis luctuosa</i> (Burmeister, 1839)			110.0
Species richness (S)	4	5	5
Abundance (mean, individuals/m ²)	1462.0	1994.0	147.9

axes were 0.75 and 0.52, explaining 45.7% of the relationships between the species and environmental variables. Species–environment ordination was statistically significant according to the Monte Carlo permutation test (first

(individuals per m²), **c** Shannon diversity index. No statistically significant differences between the pools of the three rivers were found (One-way ANOVA or Kruskal–Wallis *H* test, $p > 0.05$)

axis: F -ratio = 4.44, $p = 0.004$; overall: trace = 1.97, $F = 3.73$, $p = 0.002$). Distance from source ($R = -0.72$), water temperature ($R = -0.65$) and conductivity ($R = -0.50$) were associated with the first axis and concentration of nitrates in water with the second axis ($R = 0.48$), indicating that these were the most important parameters explaining mayfly assemblage patterns in temporary pools of studied intermittent rivers (Fig. 7).

At the river scale, mayfly abundance in sampled pools significantly increased with the distance from the river source in the Čikola River (Spearman's correlation coefficient, $R = 0.89$, $p < 0.01$), while it significantly decreased along the course of the Krčić River ($R = -0.82$, $p < 0.05$). No significant differences were recorded for other assemblage metrics (species richness, Shannon diversity) within any of the sampled intermittent rivers.

Significant results of RDA showed that mayfly functional traits differed among the pools of the three intermittent rivers (explanatory variables accounted for 65.88%; F ratio = 3.60, $p = 0.001$). The first two ordination axes (eigenvalues 0.42 and 0.18) explained 59.18% of the variation (Fig. 8). The first ordination axis is correlated mainly with water temperature ($R = 0.71$) and the second

Fig. 4 NMDS analysis of mayfly assemblages in the pools of lentic phase in three Mediterranean karst intermittent rivers (Croatia), based on Bray–Curtis similarity. Legend: 1–3, study sites along the river's course, 1 was closest and 3 furthest from the river's source. Rivers: K, Krčić; C, Čikola; MJ, Miljašić Jaruga

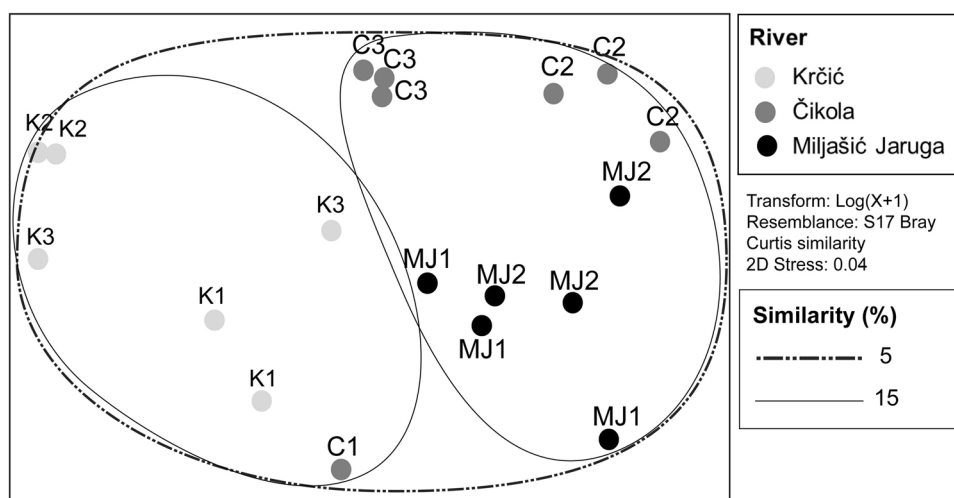
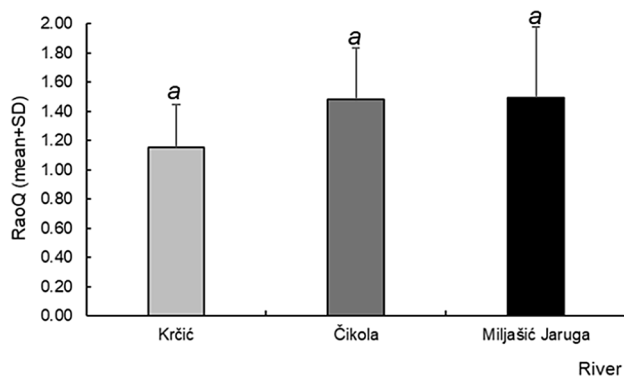


Table 5 Results of the SIMPER analysis based on mayfly assemblages in the pools of the lentic phase of three Mediterranean karst intermittent rivers (Croatia)

Species and river groups	Average abundance per pool (individuals/m ²)	Average abundance per pool (individuals/m ²)	Similarity contribution within the group (%)
Groups: Krčić & Čikola, average dissimilarity, 93.30%			
<i>Baetis rhodani</i>	Krčić 4.50	Čikola	29.77
<i>Centroptilum luteolum</i>	0.55	4.63	24.54
<i>Cloeon dipterum</i>	2.78	0.50	16.97
<i>Procloeon pennulatum</i>	0.44	3.42	16.63
<i>Caenis horaria</i>		0.90	6.50
Groups Krčić & Miljašić Jaruga, average dissimilarity, 92.17%			
<i>Baetis rhodani</i>	Krčić 4.50	Miljašić Jaruga	33.17
<i>Caenis luctuosa</i>		4.01	25.00
<i>Cloeon dipterum</i>	2.78	1.75	19.05
<i>Centroptilum luteolum</i>	0.55	2.70	17.23
Groups Čikola & Miljašić Jaruga, average dissimilarity, 73.77%			
<i>Caenis luctuosa</i>	Čikola	Miljašić Jaruga	
		4.01	28.21
<i>Centroptilum luteolum</i>	4.63	2.70	20.97
<i>Procloeon pennulatum</i>	3.42	0.47	20.46
<i>Cloeon dipterum</i>	0.50	1.75	12.79
<i>Caenis horaria</i>	0.90		8.04

**Fig. 5** Functional diversity of mayfly assemblages in the pools of the lentic phase in three Mediterranean karst intermittent rivers (Croatia) shown as mean and standard deviation (SD). No statistically significant differences between the pools of the three rivers were found (Kruskal–Wallis H test, $p > 0.05$)

with chemical oxygen demand ($R = -0.47$), indicating that these were the most important parameters explaining mayfly functional traits in temporary pools of studied intermittent rivers (Fig. 8).

Discussion

Influence of environmental variables on mayfly assemblages in temporary pools of intermittent rivers

With a total of eight recorded mayfly species and four or five species per river, this study shows that the harsh environment of temporary pools formed during the lentic phase of the Mediterranean karst intermittent rivers supports species-poor mayfly assemblages. Low mayfly diversity is not surprising considering their habitat preferences, with most species occurring in the upper reaches of fast flowing perennial streams and rivers, while much lower diversity is typical for lentic habitats, especially temporary ones (Buffagni et al. 2009, 2023; Bauernfeind and Soldán 2012; Vilenica et al. 2015). In addition, since mayflies inhabiting temporary habitats have fast life cycles (Řezníčková et al. 2010), some species could have already emerged at the time of our sampling. Many studies showed that macroinvertebrate diversity tends to be lower in intermittent than in perennial freshwater habitats (Soria et al. 2017; Stubbington et al. 2017a, b). Moreover, with the alteration of hydrological phases in intermittent rivers, a turnover of aquatic macroinvertebrate communities takes place. Namely, with water flow cessation, rheophil taxa such as Ephemeroptera are predominantly replaced with generalist taxa typical for lentic pools, such as Odonata, Coleoptera, Hemiptera, and Gastropoda (Boulton

Table 6 Differences in mayfly functional traits among the pools of the three Mediterranean karst intermittent rivers (Croatia) (Kruskal–Wallis H test, $d.f. = 2$). Significant results ($p < 0.05$) are reported in bold

Functional trait group	Functional trait	Statistical test	p -Value	Comparison
Current preference	Limnophil	$H = 2.91$	$p > 0.05$	
	Rheophil to limnophil	$H = 9.56$	$p < 0.01$	Miljašić Jaruga > Čikola, Krčić
	Rheophil	$H = 15.70$	$p < 0.001$	Krčić > Miljašić Jaruga, Čikola
Lateral connectivity preference	Plesiopotamon	$H = 9.78$	$p < 0.01$	Miljašić Jaruga > Krčić
	Palaeopotamon	$H = 9.92$	$p < 0.01$	Miljašić Jaruga > Krčić
	Temporary waterbodies	$H = 6.51$	$p < 0.05$	Čikola > Krčić
	Eupotamon	$H = 5.37$	$p > 0.05$	
	Parapotamon	$H = 5.77$	$p > 0.05$	
Trophic structure	Gatherers	$H = 7.04$	$p < 0.05$	Miljašić Jaruga > Čikola, Krčić
	Grazers	$H = 7.04$	$p < 0.05$	Čikola, Krčić > Miljašić Jaruga
Life cycle type	Bivoltine	$H = 4.66$	$p > 0.05$	
	Trivoltine	$H = 17.06$	$p < 0.001$	Krčić > Miljašić Jaruga, Čikola
	Flexible	$H = 10.82$	$p < 0.01$	Miljašić Jaruga > Krčić

2003; Bonada et al. 2006, 2008; Stubbington et al. 2009). The ponding phase of intermittent rivers increases the overall biotic diversity of such habitats, as lentic macroinvertebrate specialists colonize the instream temporary pools (Stubbington et al. 2017c).

Although species richness was comparable among the pools of the studied intermittent rivers, species composition was mostly river specific. This is most probably the result of the interplay of river and pool hydromorphology, abiotic factors, and microhabitat composition. Our results showed that water temperature, conductivity, nitrate concentration, and chemical oxygen demand were the most important physicochemical water properties shaping mayfly taxonomic and functional diversity in the studied temporary pools. Rather high values of water temperature and conductivity most probably led to overall low mayfly assemblage metrics. Furthermore, in-water nitrate concentration was highest in the pools of the Miljašić Jaruga River, most probably originating from the surrounding agricultural fields (see also Vilenica et al. 2022, 2023), and likely resulting in low population densities of mayfly nymphs due to their sensitivity to water pollution (Beketov 2004). The highest chemical oxygen demand, recorded in the Čikola River, was most likely a result of decaying organic matter (abundant macrophytes, see Vilenica et al. 2022) and wastewater inflow. Therefore, lower mayfly abundance was recorded in the pools with a higher share of decaying macrophytes and at pools located in the vicinity of urban areas (Drniš town).

The distance of the pools from the river source was another highly influential factor determining mayfly assemblage metrics. However, a specific pattern was not observed, i.e., with increasing distance from the source, mayfly

abundances were significantly lower in the pools along the Krčić River, while they were significantly higher in the pools along the Čikola River. More precisely, the highest abundance of mayfly nymphs was observed in the pools located closest to the Krčić river source, most probably due to the longer persistence of those pools, resulting in abundant development of eurytopic *Cloeon dipterum*, known to occur in large quantities at a habitat where it is present (Buffagni et al. 2009, 2023). The opposite trend was observed at the Čikola River, with the highest abundance recorded in the pools located in the canyon section of this river (i.e., furthest from the river source). At this location, the riverbed was covered with megalithal, overgrown with algae in the temporary pools, which could have provided a food resource for the partial grazers recorded there, *Procloeon pennulatum* and *Centroptilum luteolum* (Buffagni et al. 2009, 2023; Bauernfeind and Soldán 2012). Moreover, no vertebrate predators were observed in these pools, in contrast to the pools located upstream. Alternatively, high mayfly abundance in some temporary pools could be the result of the reduction of available habitat during the process of pool drying out (Storey and Quinn 2008).

Our results corroborate the findings of previous studies showing that temporary pools generally have high and variable water temperatures. This results in increased concentration of nutrients and conductivity, and decreased concentration of available oxygen, which often represent limiting factors for the occurrence of aquatic macroinvertebrates in the lentic phase of intermittent rivers (Quinn et al. 1994; Lake 2011; Drummond et al. 2015; Stubbington et al. 2017b).

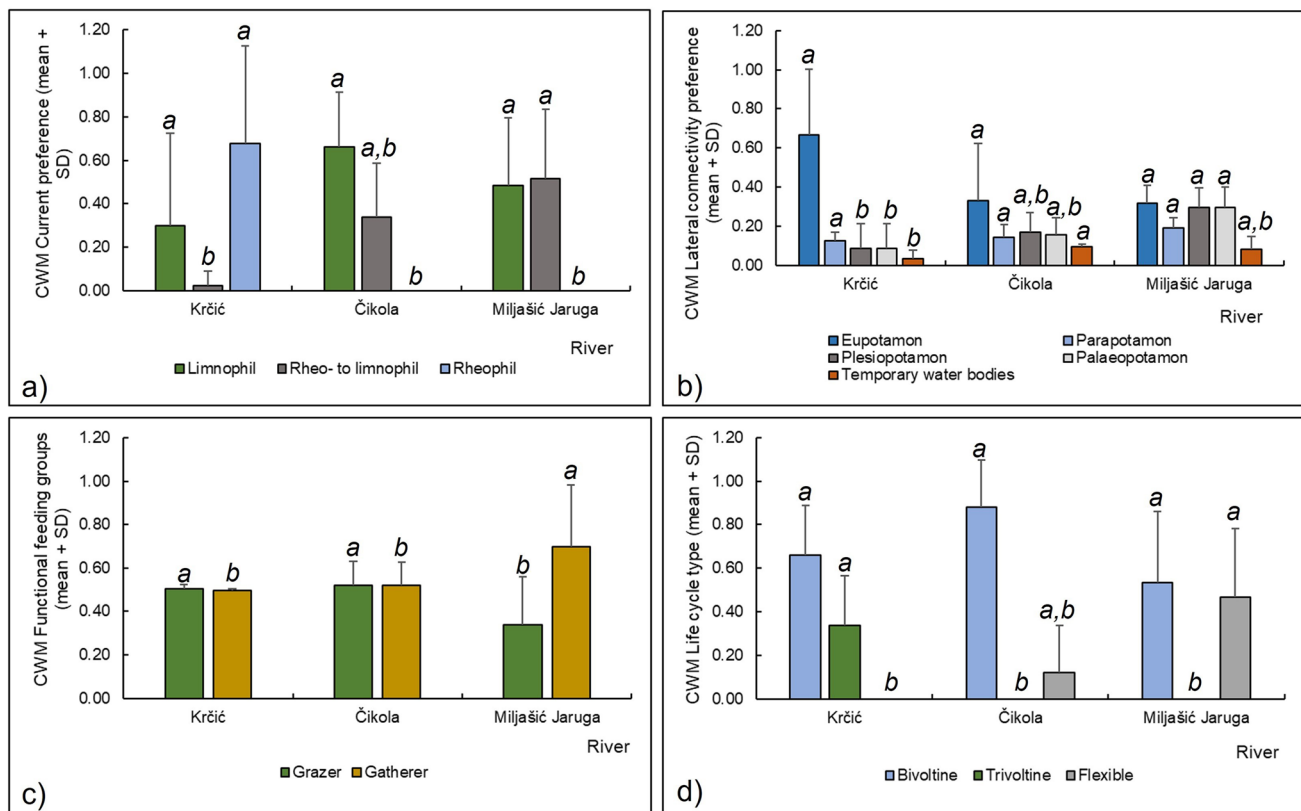


Fig. 6 Mayfly functional traits: **a** current preference, **b** lateral connectivity preference, **c** trophic structure (functional feeding groups), and **d** life cycle type of the constituent species in the pools of three Mediterranean karst intermittent rivers (Croatia). Different letters

denote statistically significant differences of a single functional trait among the temporary pools of the studied rivers (Kruskal–Wallis H test with multiple comparisons post hoc test, for significant results $p < 0.05$). CWM, community weighted mean values

Mayfly functional diversity in temporary pools of intermittent rivers

Mayfly functional diversity was comparable among the pools of the investigated intermittent rivers. It was primarily characterized by traits that allowed species to thrive in temporary freshwater habitats, where specific adaptations and strategies are crucial to enable organisms' survival in habitats with variable and extreme environmental conditions. Functional traits such as: (1) eurytopic occurrence, (2) low rheophily (low affinity to current velocity), (3) gatherer feeding group, (4) flexible life cycle, (5) polyvoltinism and (6) short life cycle (up to 1 year) are among the resilience traits characteristic for taxa inhabiting such harsh freshwater environments, enabling faster recolonization and enhanced survival during the lentic phase (Řezníčková et al. 2010; Stubbington et al. 2017b).

Mayfly habitat and water current preferences are strongly related (Buffagni et al. 2009, 2023; Bauernfeind and Soldán 2012), which is further confirmed by our data, showing prevalence of rheophil to limnophil and limnophil species in the studied assemblages. The recorded species are mainly

generalists that can occur in both lotic (with lentic sections) and lentic habitats (Buffagni et al. 2009, 2023; Vilenica et al. 2015), which enables them to adapt to a hydrologically altering environment. However, in line with the recorded differences in microhabitat composition, physicochemical water properties, and morphology of each riverbed (see Vilenica et al. 2022), some differences in above-mentioned functional traits were recorded among the pools of the three intermittent rivers. Predominantly lentic assemblages were present in the pools of the Miljašić Jaruga and Čikola Rivers, likely due to the prevalence of fine substrate and phytal microhabitats, causing the pools to persist longer, and lowering water velocity in the lotic phase (Buffagni et al. 2009, 2023; Vilenica et al. 2023). In contrast, mayfly assemblages in the pools of the canyon-type Krčić River were more lotic and more rheophilic due to the dominance of eurytopic *Cloeon dipterum* (Buffagni et al. 2009, 2023) in the pools located in the river's upper reaches, and rhithral preferring rheophil *Baetis rhodani* in the pools along the entire course of the river.

Although rheophily and preference for lotic habitats are considered to be disadvantageous traits for survival in

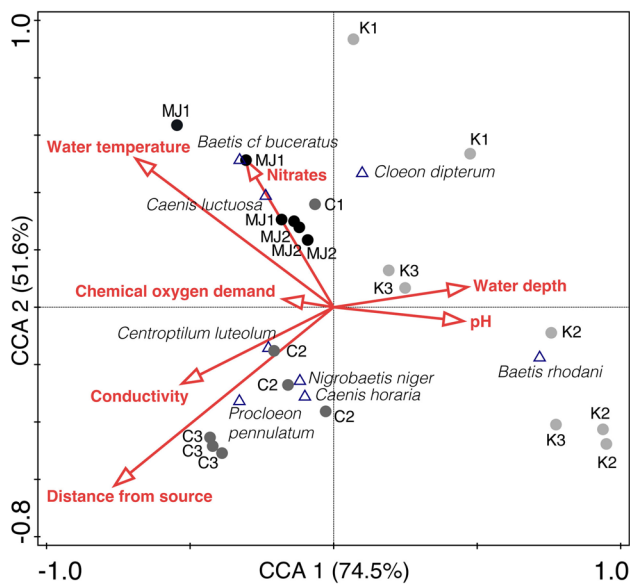


Fig. 7 Canonical correspondence analysis (CCA) triplot based on mayfly species (blue triangle symbols) and environmental variables in the pools of three Mediterranean karst intermittent rivers (Croatia). Environmental variables (red arrow symbols) measured in the pools: water depth (m), water temperature ($^{\circ}\text{C}$), pH, conductivity ($\mu\text{S}/\text{cm}$), chemical oxygen demand ($\text{mg O}_2/\text{L}$), nitrate concentration ($\text{mg N}/\text{L}$), nitrite concentration ($\text{mg N}/\text{L}$). Legend: 1–3, study sites (circle symbols) along the river's course, 1 was closest and 3 furthest from the river's source. Rivers: K, Krčić; C, Čikola; MJ, Miljašić Jaruga

intermittent rivers (Řezníčková et al. 2010, 2013), previous studies recognized *Baetis rhodani* as one of the most successful species able to survive the hydrologically harsh environmental conditions and recolonize intermittent habitats (Řezníčková et al. 2010). The authors suggested that its traits, such as life cycle flexibility, asynchrony in embryogenesis, nymphal growth, and emergence, enable dry period survival of at least some nymphal cohorts that can recolonize the intermittent stream after rewetting. Nevertheless, since *Baetis rhodani* is a species complex (Williams et al. 2006); its habitat choice in intermittent habitats should be further examined in future studies, preferably also using molecular methods.

The trophic structure of mayfly assemblages consisted of gatherers and mixed type grazer–gatherers, functional feeding groups that are considered advantageous and indifferent manifestation for drought survival, respectively (Řezníčková et al. 2010). The differences in mayfly assemblage metrics among the studied pools were evident in their trophic structure in relation to microhabitat composition, and consequently to the availability of different food resources. The pools in the Miljašić Jaruga River with a higher share of fine sediments harbored a higher proportion of gatherers that predominantly feed on fine particulate organic matter (Buffagni et al. 2009, 2023). The pools in the Krčić River

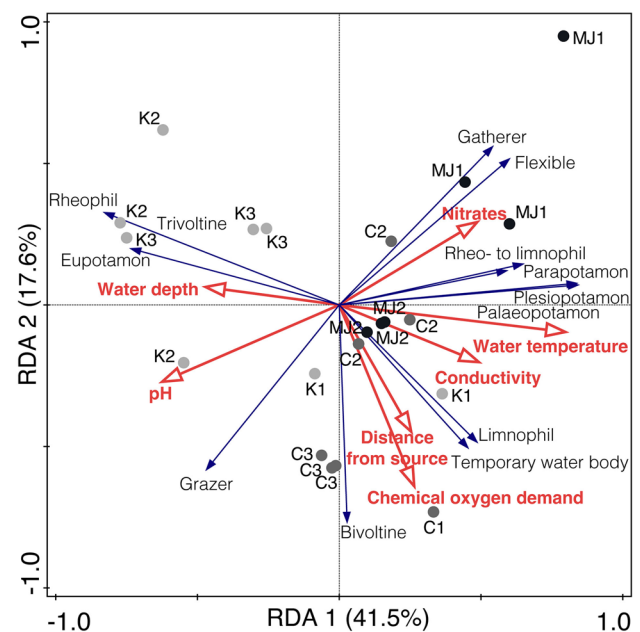


Fig. 8 Redundancy analysis (RDA) ordination triplot showing the relationships between mayfly functional traits (blue arrow symbols) and environmental variables (red arrow symbols) in the pools of three Mediterranean karst intermittent rivers (Croatia). Legend: 1–3, study sites (circle symbols) along the river's course, 1 was closest and 3 furthest from the river's source. Rivers: K, Krčić; C, Čikola; MJ, Miljašić Jaruga

and in the Čikola River canyon section, on the contrary, were dominated by lithal and phytal microhabitats (mosses or angiosperms), supporting a higher proportion of partial grazers, mostly feeding on algae and biofilm (Buffagni et al. 2009, 2023).

Other traits beneficial in drying environments, such as fast, bivoltine, or flexible life cycles (i.e., flexible number of generations per year), with nymphal development occurring in several cohorts throughout year, or predominantly during the summer (Buffagni et al. 2009, 2023; Bauernfeind and Soldán 2012), might have enabled survival and quick life cycle completion of the recorded mayflies in the short-lasting environment of the temporary pools in the studied intermittent rivers (Řezníčková et al. 2010).

Conclusions

This study showed that small and shallow temporary pools of Mediterranean karst intermittent rivers have low suitability as mayfly habitats, since a rather low number of predominantly generalist species was recorded. Taxonomic and functional diversity of mayfly assemblages in the pools were primarily influenced by distance from the river source and physicochemical water properties related to morphology, substrate composition of rivers and their pools, as well as the

level of anthropogenic impact present. However, our results showed that in such harsh environments, even some species considered rare in Croatian freshwater habitats (*Procladius pennulatus* and *Nigrobaetis niger*) (Vilenica et al. 2015) were able to survive. This indicates that the conservation value of Mediterranean intermittent rivers does not rest upon a single hydrological phase, but upon an interchange of different phases. Our results can contribute to the development of conservation measures for endangered karst intermittent rivers in the Mediterranean and the biota they support.

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Author contributions MV and AB conceived the ideas and designed methodology; MV, FR, LR, MR, RMK, and AB collected the data; MV and AB analyzed and visualized the data; and MV and AB led the writing of the manuscript. All authors have read and agreed to the published version of the manuscript.

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Data availability Data are available from the corresponding authors upon request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethics approval statement This research was conducted following the ethical guidelines of the Croatian Ministry of Economy and Sustainable Development (permit no. UP/I-612-07/20-48/210; 517-05-1-1-21-5).

References

- Amoros C, Roux AL, Reygrobellet JL, Bravard JP, Pautou G (1987) A method for applied ecological studies of fluvial hydrosystems. *Regul Rivers Res Manag* 1:17–36. <https://doi.org/10.1002/rrr.3450010104>
- Baskin Y (1994) California's ephemeral vernal pools may be a good model for speciation. *Bioscience* 44:384–385. <https://doi.org/10.2307/1312360>
- Bauernfeind E, Humpesch U (2010) Die Eintagsfliegen Zentraleuropas—Bestimmung und Ökologie. Naturhistorisches Museum, Vienna, Austria
- Bauernfeind E, Moog O (2000) Mayflies (Insecta: Ephemeroptera) and the assessment of ecological integrity: a methodological approach. *Hydrobiologia* 422:71–83. <https://doi.org/10.1023/A:1017090504518>
- Bauernfeind E, Soldán T (2012) The mayflies of Europe (Ephemeroptera). Apollo Books, Ollerup
- Beketov M (2004) Different sensitivity of mayflies (Insecta, Ephemeroptera) to ammonia, nitrite and nitrate: linkage between experimental and observational data. *Hydrobiologia* 528:209–216. <https://doi.org/10.1007/s10750-004-2346-4>
- Blaustein L, Schwartz SS (2001) Why study ecology in temporary pools? *Isr J Zool* 47:303–312
- Boersma KS, Bogan MT, Henrichs BA, Lytle DA (2014) Invertebrate assemblages of pools in arid-land stream have high functional redundancy and are resistant to severe drying. *Freshw Biol* 59:491–501. <https://doi.org/10.1111/fwb.12280>
- Bogan MT, Boersma KS, Lytle DA (2015) Resistance and resilience of invertebrate communities to seasonal and suprasedational drought in arid-land headwater streams. *Freshw Biol* 60:2547–2558. <https://doi.org/10.1111/fwb.12522>
- Bonacci O (1985) Hydrological investigations of the Dinaric karst at the Krčić catchment and the river Krka springs (Yugoslavia). *J Hydrol* 82:317–326. [https://doi.org/10.1016/0022-1694\(85\)90024-1](https://doi.org/10.1016/0022-1694(85)90024-1)
- Bonacci O, Jukić D, Ljubenkov I (2006) Definition of catchment area in Karst: case of the rivers Krčić and Krka, Croatia. *Hydrol Sci J* 51:682–699. <https://doi.org/10.1623/hysj.51.4.682>
- Bonacci O, Terzić J, Roje-Bonacci T, Frangen T (2019) An intermittent Karst river: the case of the Čikola river (Dinaric Karst, Croatia). *Water* 11:2415. <https://doi.org/10.3390/w11112415>
- Bonacci O (2014) Ecohydrology of karst poljes and their vulnerability. In: Sackl P, Durst R, Kotrošan D, Stumberger (eds) *Dinaric Karst Poljes - Floods for Life*. Euronatur, Radolfzell, Germany, pp 25–37
- Bonada N, Rieradevall M, Prat N, Resh VH (2006) Benthic macroinvertebrate assemblages and macrohabitat connectivity in Mediterranean-climate streams of northern California. *J North Am Benthol Soc* 25:32–43. [https://doi.org/10.1899/0887-3593\(2006\)25\[32:BMAAMC\]2.0.CO;2](https://doi.org/10.1899/0887-3593(2006)25[32:BMAAMC]2.0.CO;2)
- Bonada N, Rieradevall M, Dallas H, Davis J, Day J, Figueroa R, Resh VH, Prat N (2008) Multi-scale assessment of macroinvertebrate richness and composition in Mediterranean-climate rivers. *Freshw Biol* 53:772–788. <https://doi.org/10.1111/j.1365-2427.2007.01940.x>
- Borg Galea A, Sadler JP, Hannah DM, Detry T, Dugdale SJ (2019) Mediterranean intermittent rivers and ephemeral streams: challenges in monitoring complexity. *Ecohydrology* 12(8):e2149. <https://doi.org/10.1002/eco.2149>
- Boulton AJ (1989) Over-summering refuges of aquatic macroinvertebrates in two intermittent streams in central Victoria. *Trans R Soc S Aust* 113:23–34
- Boulton AJ (2003) Parallels and contrasts in the effects of drought on stream macroinvertebrate assemblages. *Freshw Biol* 48:1173–1185. <https://doi.org/10.1046/j.1365-2427.2003.01084.x>
- ter Braak CJF, Šmilauer P (2012) Canoco reference manual and user's guide: Software for ordination, Version 5.0. Micro-computer Power. Ithaca, USA
- Brittain JE, Sartori M (2003) Ephemeroptera (Mayflies). In: Resh VH, Cardé RT (eds) *Encyclopedia of Insects*. Academic Press, Amsterdam, pp 373–380
- Buffagni A, Cazzola M, Lopez-Rodriguez MJ, Alba-Tercedor J, Armanini DG (2009) Volume 3 - Ephemeroptera. In: Schmidt-Kloiber A, Hering D (eds) *Distribution and Ecological Preferences of European Freshwater Organisms*. Pensoft Publishers, Sofia-Moscow, pp 1–254
- Buffagni A, Armanini DG, Cazzola M, Alba-Tercedor J, Lopez-Rodriguez MJ, Murphy J, Sandin L, Schmidt-Kloiber A (2023) Dataset “Ephemeroptera”. www.freshwaterecology.info - the taxa and autecology database for freshwater organisms, version 8.0. Accessed 20 September 2023
- Chester ET, Robson BJ (2011) Drought refuges, spatial scale and recolonisation by invertebrates in non-perennial streams. *Freshw Biol* 56:2094–2104. <https://doi.org/10.1111/j.1365-2427.2011.02644.x>

- Clarke KR, Gorley RN (2006) Primer V6: User Manual/Tutorial. Primer-E, Plymouth
- Courtwright J, May CL (2013) Importance of terrestrial subsidies for native brook trout in Appalachian intermittent streams. *Freshw Biol* 58:2423–2438. <https://doi.org/10.1111/fwb.12221>
- Datry T, Larned ST, Fritz KM, Bogan MT, Wood PJ, Meyer EI, Santos AN (2014) Broad-scale patterns of invertebrate richness and community composition in temporary rivers: effects of flow intermittence. *Ecography* 37:94–104. <https://doi.org/10.1111/j.1600-0587.2013.00287.x>
- Datry T, Moya N, Zubieta J, Oberdorff T (2016) Determinants of local and regional communities in intermittent and perennial headwaters of the Bolivian Amazon. *Freshw Biol* 61:1335–1349. <https://doi.org/10.1111/fwb.12706>
- Datry T, Corti R, Heino J, Hugueny B, Rolls RJ, Ruhí A (2017) Habitat fragmentation and metapopulation, metacommunity, and metaecosystem dynamics in intermittent rivers and ephemeral streams. In: Datry T, Bonada N, Boulton A (eds) *Intermittent Rivers and Ephemeral Streams: Ecology and Management*. Academic Press, London, pp 377–403
- Datry T, Boulton AJ, Bonada N, Fritz K, Leigh C, Sauquet E, Tockner K, Hugueny B, Dahm CN (2018) Flow intermittence and ecosystem services in rivers of the Anthropocene. *J Appl Ecol* 55:353–364. <https://doi.org/10.1111/1365-2664.12941>
- Drummond LR, McIntosh AR, Larned ST (2015) Invertebrate community dynamics and insect emergence in response to pool drying in a temporary river. *Freshw Biol* 60:1596–1612. <https://doi.org/10.1111/fwb.12591>
- Državni Hidrometeorološki Zavod Hrvatske. <https://hidro.dhz.hr/>. Accessed 16 January 2023)
- Elliott JM, Humpesch UH, Macan TT (1988) Larvae of the British Ephemeroptera: a key with ecological notes. Freshwater Biological Association, Ambleside
- Espinosa ACE, Cunha EJ, Shimano Y, Rolim S, Mioli L, Juen L, Dunck B (2023) Functional diversity of mayflies (Ephemeroptera, Insecta) in streams in mining areas located in the Eastern Amazon. *Hydrobiologia* 850:929–945. <https://doi.org/10.1007/s10750-022-05134-x>
- Ferro ML, Sites RW (2007) The Ephemeroptera, Plecoptera, and Trichoptera of Missouri State Parks, with notes on biomonitoring, mesohabitat associations, and distribution. *J Kans Entomol Soc* 80:105–129. [https://doi.org/10.2317/0022-8567\(2007\)80\[105:TEPATO\]2.0.CO;2](https://doi.org/10.2317/0022-8567(2007)80[105:TEPATO]2.0.CO;2)
- Greenberg AE (1992) *Standard Methods for the Examination of Water and Wastewater*, 18th ed. American Public Health Association (APHA), Washington DC, USA
- Hill MJ, Milner VS (2018) Ponding in intermittent streams: a refuge for lotic taxa and a habitat for newly colonising taxa? *Sci Total Environ* 628(629):1308–1316. <https://doi.org/10.1016/j.scitotenv.2018.02.162>
- Illies J (1978) *Limnofauna Europaea*. Gustav Fischer Verlag, Stuttgart
- Lake PS (2011) *Drought and aquatic ecosystems: effects and responses*. Wiley-Blackwell, West Sussex
- Larned ST, Datry T, Arscott DB, Tockner K (2010) Emerging concepts in temporary-river ecology. *Freshw Biol* 55:717–738. <https://doi.org/10.1111/j.1365-2427.2009.02322.x>
- Leigh C, Boulton AJ, Courtwright JL, Fritz K, May CL, Walker RH, Datry T (2016) Ecological research and management of intermittent rivers: an historical review and future directions. *Freshw Biol* 61:1181–1199. <https://doi.org/10.1111/fwb.12646>
- Lenat DR, Penrose DL (1996) History of the EPT taxa richness metric. *J North Am Benthol Soc* 13:305–306
- Lynch RJ, Bunn SE, Catterall CP (2002) Adult aquatic insects: potential contributors to riparian food webs in Australia's wet-dry tropics. *Austral Ecol* 27:515–526. <https://doi.org/10.1046/j.1442-9993.2002.01208.x>
- Miliša M, Mihaljević Z, Pozojević I (2020) Report on fitting a macroinvertebrate classification method with the results of the completed intercalibration of the MED GIG (R-M5). Technical report. University of Zagreb, Faculty of Science, Zagreb, Croatia
- Paltridge RM, Dostine PL, Humphrey CL, Boulton AJ (1997) Macroinvertebrate recolonization after rewetting of a tropical seasonally-flowing stream (Magela Creek, Northern Territory, Australia). *Mar Freshw Res* 48:633–645. <https://doi.org/10.1071/MF97059>
- Quinn JM, Steele GL, Hickey CW, Vickers ML (1994) Upper thermal tolerances of twelve New Zealand stream invertebrate species. *New Zeal J Mar Fresh* 28:391–397. <https://doi.org/10.1080/00288330.1994.9516629>
- Řezníčková P, Soldán T, Pařil P, Zahrádková S (2010) Comparison of mayfly (Ephemeroptera) taxocenes of permanent and intermittent Central European small streams via species traits. *Biologia* 65:720–729. <https://doi.org/10.2478/s11756-010-0067-x>
- Řezníčková P, Tajmrová L, Pařil P, Zahrádková S (2013) Effects of drought on the composition and structure of benthic macroinvertebrate assemblages – a case study. *Acta Univ Agric Silv Mendel Brun* 61:1853–1865. <https://doi.org/10.11118/actaun201361061853>
- Richardson WB (1990) A comparison of detritus processing between permanent and intermittent headwater streams. *J Freshw Ecol* 5:341–357. <https://doi.org/10.1080/02705060.1990.9665247>
- Ricotta C, Moretti M (2011) CWM and Rao's quadratic diversity: a unified framework for functional ecology. *Oecologia* 167:181–188. <https://doi.org/10.1007/s00442-011-1965-5>
- Robson B, Chester E, Austin C (2011) Why life history information matters: drought refuges and macroinvertebrate persistence in non-perennial streams subject to a drier climate. *Mar Freshw Res* 62:801–810. <https://doi.org/10.1071/MF10062>
- Schmedtje U, Colling M (1996) Ökologische Typisierung der aquatischen Makrofauna. Informationsberichte des Bayerischen Landesamtes für Wasserwirtschaft 4/96, München
- Schmidt-Kloiber A, Hering D (2015) www.freshwaterecology.info—An online tool that unifies, standardises and codifies more than 20,000 European freshwater organisms and their ecological preferences. *Ecol Indic* 53:271–282. <https://doi.org/10.1016/j.ecoli.2015.02.007>
- Schriever TA, Lytle DA (2016) Convergent diversity and trait composition in temporary streams and ponds. *Ecosphere* 7:e01350. <https://doi.org/10.1002/ecs2.1350>
- Šegota T, Filipčić A (2003) Köppenova podjela klima i hrvatsko nazivlje. *Geoadria* 8:17–37. <https://doi.org/10.15291/geoadria.93>
- Soria M, Leigh C, Datry T, Bini LM, Bonada N (2017) Biodiversity in perennial and intermittent rivers: a meta-analysis. *Oikos* 126(8):1078–1089. <https://doi.org/10.1111/oik.04118>
- StatSoft (2010) STATISTICA 10.0 for Windows. StatSoft Inc., Tulsa
- Steward AL, Von Schiller D, Tockner K, Marshall JC, Bunn SE (2012) When the river runs dry: human and ecological values of dry riverbeds. *Front Ecol Environ* 10:202–209. <https://doi.org/10.1890/110136>
- Storey RG, Quinn JM (2008) Composition and temporal changes in macroinvertebrate communities of intermittent streams in Hawke's Bay, New Zealand. *New Zeal J Mar Fresh* 42(1):109–125. <https://doi.org/10.1080/00288330809509941>
- Stubbington R (2012) The hyporheic zone as an invertebrate refuge: a review of variability in space, time, taxa and behaviour. *Mar Freshw Res* 63:293–311. <https://doi.org/10.1071/MF11196>
- Stubbington R, Greenwood AM, Wood PJ, Armitage PD, Gunn J, Robertson AL (2009) The response of perennial and temporary headwater stream invertebrate communities to hydrological extremes. *Hydrobiologia* 630:299–312. <https://doi.org/10.1007/s10750-009-9823-8>
- Stubbington R, England J, Sefton C, Wood PJ (2017a) Temporary streams in temperate zones: recognizing, monitoring and restoring

- transitional aquatic-terrestrial ecosystems. Wiley Interdiscip Rev Water 4:e1223. <https://doi.org/10.1002/wat2.1223>
- Stubbington R, Bogan MT, Bonada N, Boulton AJ, Datry T, Leigh C, Vander Vorste R (2017b) Chapter 4.3 - The Biota of Intermittent Rivers and Ephemeral Streams: Aquatic Invertebrates. In: Datry T, Bonada N, Boulton A (eds) Intermittent Rivers and Ephemeral Streams. Academic Press, London, pp 217–243
- Stubbington R, England J, Wood PJ, Sefton CEM (2017c) Temporary streams in temperate zones: recognising, monitoring and restoring transitional aquatic-terrestrial ecosystems. Wiley Interdiscip Rev Water 4:e1223. <https://doi.org/10.1002/wat2.1223>
- Tilman D (2001) Functional diversity. In: Levin SA (ed) Encyclopedia of Biodiversity. Academic Press, San Diego, pp 109–120
- Vilenica M, Gattolliat J-L, Mihaljević Z, Sartori M (2015) Croatian mayflies (Insecta, Ephemeroptera): species diversity and distribution patterns. ZooKeys 523:99–127. <https://doi.org/10.3897/zookeys.523.6100>
- Vilenica M, Rebrina F, Matonićkin Kepčija R, Šegota V, Rumišek M, Ružanović L, Brigić A (2022) Aquatic macrophyte vegetation promotes taxonomic and functional diversity of odonata assemblages in intermittent karst rivers in the mediterranean. Diversity 14:31. <https://doi.org/10.3390/d14010031>
- Vilenica M, Rebrina F, Ružanović L, Rumišek M, Matonićkin Kepčija R, Brigić A (2023) Are intermittent rivers in the Karst Mediterranean Region of the Balkans Suitable as Mayfly Habitats? Diversity 15(2):155. <https://doi.org/10.3390/d15020155>
- Villéger S, Mason NWH, Moullot D (2008) New multidimensional functional diversity indices for a multifaceted framework in functional ecology. Ecology 89:2290–2301
- Waltz RD, Mccafferty WP, Thomas A (1994) Systematics of *Alainites* n. Gen., *Dipheter*, *Indobaetis*, *Nigrobaetis* n. Stat., and *Takobia* n. Stat. (Ephemeroptera, Baetidae). Bull Soc Hist Nat Toulouse 130:33–36
- Williams DD (2006) The Biology of Temporary Waters. Oxford University Press, Oxford
- Williams P, Biggs J, Fox G, Nicolet P, Whitfield M (2001) History, origins and importance of temporary ponds. Freshwater Forum 17:7–15
- Williams HC, Ormerod SJ, Bruford MW (2006) Molecular systematics and phylogeography of the cryptic species complex *Baetis rhodani* (Ephemeroptera, Baetidae). Mol Phylogenet Evol 40(2):370–382. <https://doi.org/10.1016/j.ympev.2006.03.004>
- Zacharias I, Dimitriou E, Dekker A, Dorsman E (2007) Overview of temporary ponds in the Mediterranean region: threats, management and conservation issues. J Environ Biol 28:1–9
- Zaninović K, Gajić-Čapka M, Perčec Tadić M, Vučetić M, Milković J, Bajić A, Cindrić K, Cvitan L, Katušin Z, Kaučić D, Likso T, Lončar E, Lončar Ž, Mihajlović D, Pandžić K, Patarčić M, Srnc L, Vučetić V (2008) Climate atlas of Croatia 1961–1990, 1971–2000. State Hydrometeorological Institute, Zagreb

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